nature portfolio

Peer Review File

High-accuracy retinal age prediction via fundus-based multitask learning reveals the effect of systemic disease

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This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
Peer Review Report
Summary:
This study presents a novel multitask learning framework that significantly improves retinal age prediction accuracy (MAE 2.78 years) by integrating fundus images with HbA1c data, outperforming existing benchmarks while revealing clinically meaningful associations between retinal age gaps and systemic diseases like diabetes (+1.31 years) and stroke (+0.75 years). The study demonstrating the retina's potential as a non-invasive window to systemic health, though future validation in larger, more diverse cohorts remains needed to confirm broader applicability.

Major Concerns:

1. External Validation Sample Size Concerns
The external validation cohort comprising only 51 participants represents a significant methodological limitation. This small sample size is insufficient to comprehensively evaluate model performance and generalization. Additionally, while Table 1 currently presents statistical comparisons between groups A (training) and B (internal validation), it would strengthen the analysis to include formal statistical testing between groups B and C (internal vs. external validation) to better characterize dataset heterogeneity.
2. Model Selection Methodology Clarification Needed
The manuscript requires greater transparency regarding the model selection process. Critical details are missing about: (1) the sample size and distribution characteristics used for model comparisons; (2) the selection criteria and basis; and (3) formal statistical testing between candidate models. The authors should supplement the methods section with: a description of the comparative analysis dataset; explicit selection criteria; p-values for inter-model performance differences. This will enable readers to properly evaluate the robustness of the chosen multitask learning approach versus alternatives.
3. Discussion Section Requires Strengthening
Several paragraphs in the discussion (particularly paragraphs 2-4) contain unsupported assertions and lack appropriate citations. The current version occasionally overinterprets results without sufficient empirical support or literature references, which diminishes the section's scientific rigor.

Reviewer #2

(Remarks to the Author)
Ninomiya et al. trained an ensemble multitask-learning model that integrates fundus photographs with HbA1c to predict biological age. The model achieved a mean absolute error (MAE) of 2.78 years in internal validation and 3.74 years in external validation. The authors also reported that the retinal age gap (the difference between retinal-predicted age and chronological age) was significantly higher in patients with cardiometabolic diseases, such as diabetes, cardiac disease, or stroke, even after adjusting for age and sex. The novelty of this approach, combined with the use of different datasets to validate robustness, are important strengths of the study. However, I have several suggestions that could further improve the manuscript.

Abstract
• Consider adding a definition of retinal age early in the abstract, as readers may not be familiar with this concept.
• The phrase accelerated biological ageing may be misleading; please revise to biological ageing to ensure accuracy.
• For the additional 7,193 images, clarify whether this cohort was disease-free.
• The statement about outperforming a public benchmark is confusing. Please specify the source of the benchmark and consider moving this point to the Discussion.
• Clarify why the standard deviation across ensemble predictions was used as an internal confidence score. What was this for?
• Only age and sex were adjusted for in the systemic disease cohort; potential confounding factors such as comorbidities and medication use should be acknowledged, as they may bias the results.

Introduction
• The statement “Retinal imaging techniques … provide insights into the health of the eye and systemic circulation” requires citations. Suggested references: Oculomics: Current Concepts and Evidence / Oculomics: A Crusade Against the Four Horsemen of Chronic Disease
• The section on AI-based disease detection should cite landmark studies in the field, e.g.: Development and Validation of a Deep Learning Algorithm for Detection of Diabetic Retinopathy in Retinal Fundus Photographs / Development and Validation of a Deep Learning System for Diabetic Retinopathy and Related Eye Diseases Using Retinal Images From Multiethnic Populations With Diabetes / Efficacy of a Deep Learning System for Detecting Glaucomatous Optic Neuropathy Based on Color Fundus Photographs
• The concept of the retinal age gap (RAG) and its association with cardiovascular disease should be supported by key references, such as: Retinal Age Gap as a Predictive Biomarker of Stroke Risk / Association of Retinal Age Gap With Arterial Stiffness and Incident Cardiovascular Disease
• For statements about the predictive potential of retinal age, please include landmark studies: Predictive Potential of Retina-Based Biological Age in Assessing Chronic Obstructive Pulmonary Disease Risk / Retinal Photograph-Based Deep Learning Predicts Biological Age and Stratifies Morbidity and Mortality Risk / Application of a Deep-Learning Marker for Morbidity and Mortality Prediction Derived From Retinal Photographs / Association of Retinal Age Gap and Risk of Kidney Failure: A UK Biobank Study / Retinal Age Gap as a Predictive Biomarker of Future Risk of Parkinson’s Disease
• The scoping review on retinal age estimation (2022–2023) should cite: Ocular Ageing Biomarkers and Their Clinical Utility: A Review / A Cross-Population Study of Retinal Aging Biomarkers With Longitudinal Pre-Training and Label Distribution Learning
• The statement that HbA1c was included as an auxiliary signal but only fundus photographs are required for deployment needs clarification. Why was only one systemic biomarker chosen for model development?
• Please elaborate on how model performance was evaluated in the MTL model. For example, in the case of MTL with two outputs (e.g., age and another variable), only one MAE is reported. Could you clarify whether this MAE refers to age only, or how the results across outputs were integrated into a single performance measure?

Results
• Please include the age range of participants in the each dataset.
• Cross-validation may risk overfitting; consider addressing this limitation.
• The Grad-CAM results need clearer explanation and interpretation.
• The rationale for stratifying by prediction standard deviation should be explicitly stated, same as above.
• When comparing participants taking antihypertensive medications versus not, those on medications showed higher RAG. Could you clarify whether the ‘non-medication’ group includes only individuals without hypertension, or also those with hypertension who are untreated? The same clarification would be helpful for diabetes and lipid-lowering medications. Same for diabetes and lipid-lowering medications.

Discussion
• The discussion section is too long and should be more concise.
• A key limitation is the absence of prospective studies validating the predictive value of the retinal age gap for systemic diseases. This should be emphasized.
• As the authors mentioned a subset of participants with three consecutive years of retinal images, but only used the earliest instance for internal validation, could you please also discuss the potential future research of longitudinal validation of the model?

Reviewer #3

(Remarks to the Author)
This manuscript presents a multitask learning (MTL) approach for retinal age prediction using fundus images and HbA1c, with ensemble refinement. However, the work lacks significant innovation and overlooks advancements in retinal AI. Key methodological gaps and insufficient validation weaken the claims of improved generalizability and clinical utility. Below, I detail major concerns.
1. The proposed MTL framework (Fig. 5b) combining age and HbA1c prediction offers marginal MAE improvements over single-task models (Table 2), yet the methodology—using systemic parameters to enhance retinal biomarkers—is well-established in prior literature. The authors fail to justify why only two tasks (age + HbA1c) were optimized, ignoring whether scaling to 3+ tasks (e.g., +blood pressure/LDL) would further improve performance or exacerbate resource demands. Given

that blood parameters are already integrated into retinal systemic disease screening (e.g., CKD/DKD, Deep-learning models for the detection and incidence prediction of chronic kidney disease and type 2 diabetes from retinal fundus images. Nat Biomed Eng 2021 & Non-invasive biopsy diagnosis of diabetic kidney disease via deep learning applied to retinal images: a population-based study. Lancet Digit Health 2025), this approach does not advance the field meaningfully. Although these studies did not utilize blood test parameters as outputs of deep learning models, it has been empirically demonstrated that such parameters can enhance the performance of retinal models in machine learning approaches (e.g., Random Forest and XGBoost).
2. The opinion of foundation models (e.g., RETFound) in the Discussion (L219-221) is misleading. While training such models requires resources, fine-tuning them for specific tasks (like retinal age prediction in this case) is computationally efficient and requires minimal annotated data—a fact well-documented in Zhou et al., 2023. The authors must compare their MTL ensemble against a fine-tuned RETFound baseline to validate claims of superior generalizability across datasets (L255-260).
3. The authors should note the recent development of multimodal foundation models in medicine, particularly in ophthalmology (A multimodal visual–language foundation model for computational ophthalmology. npj Digit Med 2025 & An eyecare foundation model for clinical assistance: a randomized controlled trial. Nat Med 2025), which integrate retinal images with clinical data into a unified framework. This approach can effectively address the aforementioned issues.
4. As mentioned in L340–345 and L359–363 regarding limitations, the homogeneous and insufficient external validation data (51 eyes) provides limited support for the conclusions. Prior retinal-age studies (e.g., ref 6, 7, 15) leveraged large-scale cohorts like UK Biobank; omitting similar validation here is a oversight.
5. Furthermore, the manuscript fails to provide novel clinical insights, as the relationship between retinal age and systemic diseases is already well-established. While the authors claim that their MTL and ensemble approach enhances predictive accuracy and improves generalisability (L84-88), the clinical impact of such "highly accurate" age predictions remains unclear and underexplored. There is no substantive evidence demonstrating improved generalisability to new data, as the claims lack support from strong baseline comparisons and extensive external validation.
6. Several technical details lack clarity. The quality control pipeline for fundus images (Fig. 4) is described superficially. Table 2 notes models used "fundus + HbA1c" or "fundus + sex" inputs, but the fusion mechanism is unspecified. The systemic disease assessment set and external set used one eye per participants. Was this eye chosen randomly, or by laterality/quality? The choice of LWBNA_Unet over a pretrained model (e.g., ResNet) is unexplained. Did the authors compare backbones? Pretrained models could improve baseline accuracy.

Version 1:

Reviewer comments:

Reviewer #2

(Remarks to the Author)
Authors had addressed all my concerns.

Reviewer #3

(Remarks to the Author)
I thank the authors for their detailed rebuttal and substantial revisions to the manuscript. The revised version represents a clear improvement over the initial submission.

I particularly appreciate the expanded external validation and the inclusion of direct head-to-head comparisons with both the JOIR benchmark and fine-tuned RETFound models. These additions address my primary concerns regarding limited validation and insufficient baselines, and they strengthen the support for the reported performance gains. In addition, the discussion of foundation models is now more balanced. The added technical details regarding quality control, feature fusion, eye selection, and model architecture improve reproducibility.

Overall, the authors have addressed the majority of my concerns, and the manuscript is significantly improved. I appreciate the authors’ responsiveness to the reviewers’ comments and hope that future work will further extend these findings through larger, more diverse cohorts and longitudinal validation.

Ninomiya T et al. COMMSMED-25-1418-T

Reply to the Reviewers

We would like to thank the Editor and all the Reviewers for their careful reading of our manuscript entitled "High-accuracy retinal age prediction via fundus-based multitask learning reveals the effect of systemic disease". We greatly appreciate their insightful and constructive comments, which have helped us substantially improve the clarity, methodological transparency, and clinical interpretation of our work.

In the revised version, we have addressed every point raised by the Reviewers. All changes made in response to the comments are incorporated into the new manuscript and can be located using the page and line numbers indicated below.

We have provided point-by-point responses to each of the Reviewer's comments. The Reviewers' comments are presented in *italics* within brackets, followed by our responses and a description of the corresponding changes in the manuscript.

Note on Table 1 and Fig. 1 cohort counts: During final data verification, we identified a discrepancy in the cohort counts reported in Table 1 and the study flowchart (Fig. 1), which arose from inadvertently using an outdated aggregation file for descriptive statistics. We have corrected Table 1, Fig. 1, and the corresponding cohort-size statements throughout the manuscript to reflect the final quality-controlled dataset used in this study. All model training and evaluation analyses were performed on the final dataset; therefore, this correction does not affect the reported performance metrics or the conclusions. We apologise for the oversight.

<table>
<tr><th colspan="2">Reviewer 1</th></tr>
<tr><th>Comments</th><th>Responses</th></tr>
<tr><td>[#1. External validation with 51 eyes is insufficient. Previous retinal age studies have relied on large external cohorts, such as UK Biobank. The very limited external-validation sample raises concerns about generalisability. Please expand or better justify the external validation strategy, and clarify how your external data relate to the JOIR model you compare against.]</td><td>Thank you very much for your valuable comment. We admit that the original external-validation cohort of 51 participants was insufficient to support strong claims about robustness and generalisability. In response, we have substantially expanded and clarified our external validation.
First, we increased the primary external-validation cohort (External Test 1) from 51 to 135 individuals by including all in-scope participants from the Sendai Open Hospital health-screening programme who met our inclusion and exclusion criteria and had at least one gradable fundus photograph. This expansion has been described in the ‘Study population and data collection’ (page 8, lines 147–152) and ‘External validation datasets’ subsections of the Methods section (page 9, lines 171–190), and summarised in the ‘Participant characteristics of each dataset’ subsection of the Results section (page 15-16, lines 327–331).
Second, to address the Reviewer’s concern about relying on a single small external cohort, we added a second, independent external-validation cohort (External Test 2) derived from the AlzEye 2018 UK screening study. We restricted this dataset to participants of Asian ethnicity and applied diagnosis-based surrogates to approximate the inclusion and exclusion criteria used for the Japanese development cohorts, thereby mitigating ethnic and clinical domain shifts as far as possible under data-governance constraints. Details have now been provided in the ‘External validation datasets’ subsection of the Methods section (page 9, lines 170–190) and in the ‘Participant characteristics of each dataset’ subsection of the Results section (page 15-16, lines 340–350).
Third, we performed direct head-to-head comparisons between our multitask learning (MTL) ensemble and the publicly available JOIR retinal age model, as well as baseline</td></tr>
</table>

	and fine-tuned RETFound-DINOv2 foundation models, in External Test 1. In this cohort, the MTL ensemble achieved an overall MAE of 3.39 ± 2.74 years, compared with 5.37 ± 3.84 years for JOIR, 4.45 ± 3.15 years for the fine-tuned RETFound-DINOv2 model, and 4.65 ± 3.31 years for the RETFound-DINOv2 baseline (all $P \leq 0.003$ for MAE comparisons with our ensemble). We then evaluated the MTL ensemble and JOIR in the much larger External Test 2 (n = 4,992 eyes), where the MTL ensemble again demonstrated a modest but statistically significant reduction in the MAE relative to JOIR (8.63 ± 6.33 vs. 9.02 ± 7.55 years; $P = 0.003$). The design of these comparisons is described in 'Machine learning methods' (page 11, lines 234–236) and 'Statistics and Reproducibility' (page 14-15, lines 313–317), and the full results are presented in Table 3 (pages 41). Finally, we clarified the relationship between our data and JOIR. We have now explained that JOIR is treated as a benchmark model evaluated in our independent cohorts, rather than as a source of training data, and that our own development cohorts are entirely separate. This is stated in 'Machine learning methods' (page 11, lines 234–236) and in the Discussion where we interpret the external-validation results in the context of prior work (page 22–24, lines 473–530). We have also toned down our generalisability claims and explicitly acknowledged that, although we have now validated two independent external cohorts totalling more than 5,000 individuals, all cohorts are still Asian and future work must confirm performance in broader multiethnic populations (Discussion, page 24–25, lines 543–570).

[#2. The model-selection methodology is unclear. It is difficult to understand how Table 2, Table 3, and the final ensemble relate to each other. Please clarify the use of the reduced 5% subset, the cross-validation procedure, and how you selected the final model. Also specify which output is used to compute the metrics for multitask models.]	We greatly appreciate your comment and agree that the model-selection process needs clearer exposition. We have now substantially revised both the Methods and Results to provide a transparent description of the procedure. First, in the Methods, ‘Machine learning methods’, we now explain that we created a reduced ‘model-comparison dataset’ by randomly sampling approximately 5% of eye records from the training and internal-validation cohorts after applying all the inclusion and exclusion criteria (page 10, lines 211–215). We have emphasised that the demographic and clinical characteristics of this subset closely mirror those of the full development cohorts (Table 1). Using this reduced dataset, we trained candidate single-task learning (STL) and multitask learning (MTL) configurations with five-fold cross-validation and recorded the MAE, MSE, and MAPE for age prediction. We have explicitly stated that for MTL models, all performance metrics in Table 2 refer to the age-prediction output, whereas the auxiliary systemic-variable output (for example, HbA1c) serves as a regularising task and is not used as a primary model-selection criterion (page 11, lines 217–221). Second, we clarify that based on these comparisons, we selected the MTL configuration that uses fundus images to jointly predict age and HbA1c as the final architecture because it achieved the lowest ensemble MAE and showed a statistically significant improvement over the baseline STL model (page 11, lines 222–225). We then describe how this selected architecture was retrained on the full training cohort using five-fold cross-validation at the participant level to generate five fold-specific models, all evaluated on the same independent internal-validation cohort (n = 7,288). The ensemble prediction for each eye is the arithmetic mean of these five models (page 11, lines 227–232). Third, in the ‘Performance of single-task and multitask models’ subsection of the Results, we have revised the paragraphs describing Table 2 to make this workflow explicit

	(page 16–17, lines 352–374). We now state that Table 2 summarises the initial comparison on the reduced subset, all metrics correspond to the age output, and the MTL age + HbA1c configuration was chosen as the final model based on these results. Fourth, we clarified the statistical tests used in the model-selection stage. In ‘Statistics and reproducibility’, we now explain that we treated the five cross-validation folds as paired observations and used two-sided Wilcoxon signed-rank tests on fold-wise MAEs to compare each candidate model with the baseline STL configuration (page 14, lines 307–312). Table 2 has been updated accordingly to report ΔMAE and P-values relative to the baseline, with these details described in the table legend (Table 2, page 40, lines 874–885). Together, these revisions clarify the use of the reduced subset for initial model comparison, selection of the final MTL architecture, and how the ensemble reported in Tables 3–5 was obtained.
[#3. The Discussion does not sufficiently highlight what is novel about this work or how it advances the field beyond existing retinal age models and foundation-model approaches. The authors should clearly articulate their contribution and place it in the context of prior retinal ageing biomarkers, oculomics frameworks, and robustness/generalisation issues.]	We thank the Reviewer for this valuable suggestion. We have rewritten and expanded key parts of the Introduction and Discussion sections to more clearly position our work in the existing literature and articulate our specific contributions. In the Introduction, we now provide a more detailed overview of prior research on retinal age gap (RAG), citing studies that demonstrated associations with cardiovascular events, chronic obstructive pulmonary disease, mortality, kidney failure, and Parkinson’s disease (page 5, lines 80–84). We have also cited recent scoping reviews and summaries of ocular ageing biomarkers and cross-population retinal ageing models to highlight that current architectures may be reaching a performance ceiling and that many studies rely on pre-existing models with limited architectural transparency (page 5–6, lines 85–91). We have explicitly stated our design choices and

	conceptual contribution: rather than proposing a new definition of retinal age; we have focused on simple, well-controlled architectural modifications—a two-output MTL framework jointly predicting age and a single systemic biomarker, combined with an ensemble of fold-specific models—to improve accuracy and robustness while retaining interpretability (Introduction, page 6, lines 96–101). We have also explained why we restricted the MTL architecture to two regression outputs and ultimately selected HbA1c as the auxiliary target, emphasising interpretability, avoidance of negative transfer, and the stability of HbA1c as a metabolic marker (page 6, lines 100–109). In the Discussion section, we have correlated our findings explicitly to the broader oculomics agenda and multimodal retinal-phenotype frameworks. We discuss how RAG and related indices have been proposed as general indicators of systemic vulnerability (page 24, lines 531–534), and we interpret our systemic-disease associations in light of previous RAG studies (page 24, lines 538–542). Further, we note that our approach is complementary to multimodal frameworks such as RetiAGE, and that future research should explore combined models rather than positioning our method as a replacement (Discussion, page 26, lines 576–579). Finally, we more clearly frame our novelty as follows: (1) a transparent, carefully benchmarked MTL ensemble trained from scratch on a large health-check cohort, with age + HbA1c identified as the optimal configuration in systematic comparisons; (2) demonstration that this architecture achieves MAEs below 3 years internally and outperforms both a widely used retinal age model and state-of-the-art foundation models in external cohorts; and (3) detailed examination of how ensemble disagreement can be used as an internal confidence measure for stratifying systemic-disease associations. These points are now emphasised throughout the Discussion section (page 21–24, lines 449–

	530).
[#4. The rationale and usage of the ensemble standard deviation as an "internal confidence score" is not clearly described. Why use the standard deviation across the five fold-specific predictions, and how is it employed in the analyses?]	We greatly appreciate your valuable comment. We have now expanded both the Methods and Results sections to explain the motivation and usage of the ensemble standard deviation (SD). In the 'Machine learning methods' subsection, we now describe that after training five fold-specific MTL models, we computed the SD of their age predictions for each eye as an internal proxy for prediction uncertainty. Larger SD indicates greater disagreement among models and therefore lower confidence in the prediction (page 11, lines 230–232). We also state that this SD was used to stratify eyes into Low-SD and High-SD subgroups at the cohort-specific median in subsequent accuracy and systemic-disease analyses (page 11, lines 232–234). We have added a dedicated subsection 'Stratification by prediction standard deviation' to the Results section, where we explain that we have divided eyes into Low-SD and High-SD groups based on the median SD and that eyes with lower SD exhibited substantially lower MAE internally and in the primary external cohort (page 18, lines 401–410 and Figure 5). We also clarify that Table 6 presents systemic-disease associations for the overall cohort and for Low-SD and High-SD subgroups separately, and the table legend now defines the SD threshold and explains the rationale (Table 6, page 44, lines 929–934). We have interpreted this finding In the Discussion section by stating that stratifying eyes by ensemble disagreement can serve as a proxy for prediction confidence and may help identify subgroups in which retinal age is most informative (Discussion, page 24, lines 535–538).

[#5. The description of quality control (QC) and eye selection is superficial. Please clarify how poor-quality images were excluded, how gradability was determined, and how one eye per participant was chosen to avoid within-participant correlation.]	In the 'Study population and data collection' subsection, we now state that fundus image quality for the present study was reviewed by a board-certified ophthalmologist who routinely interprets fundus photographs acquired during health screening as part of routine clinical practice. Images were classified as poor quality and excluded if they showed incomplete visualisation of the optic disc or macula, severe motion blur, media opacity, or large artefacts obscuring the major retinal vessels (page 8, lines 160–162). We also clarify how one eye per participant was selected. For the systemic disease assessment set, when both eyes were available, the higher-quality eye was chosen; if quality was similar, one eye was selected at random (Methods, page 8, lines 150–155). For the external cohorts, we used a predefined rule that preferentially selected the right eye, and when multiple fundus examinations were available, we used the earliest examination date (Results, page 16, lines 348–350; Methods, 'External validation datasets', page 9, lines 183–186). These rules are now explicitly described so that readers can see how within-participant correlation was avoided. Additionally, we noted that automated checks were used to assist the expert review, and remaining variation in focus and illumination is acknowledged as a limitation in the Discussion section (page 24–25, lines 543–552).

[#6. The baseline comparison across cohorts is difficult to interpret, and the statistical methods for Table 1 are not clearly stated. Please clarify how you compared cohorts A–D and how you handled the fact that only summary statistics were available for External Test 2.]	We greatly appreciate the Reviewer for highlighting this. We have clarified both the statistical methodology and the presentation of baseline differences. In the 'Statistics and reproducibility' section, we now specify that for baseline characteristics (Table 1), group comparisons between the training, internal-validation, and External Test 1 cohorts (A–C) used two-tailed Mann–Whitney U tests for continuous variables and χ^2 tests for categorical variables when individual-level data were available (page 14, lines 297–300). For External Test 2 (cohort D), where only summary statistics were available, P-values for comparisons with the training cohort were estimated from means, standard deviations, and sample sizes using Welch's two-sample t-test for continuous variables and the χ^2 test for categorical variables (page 14, lines 301–303). To quantify between-cohort differences independently of sample size, we additionally calculated absolute standardised mean differences (|SMD|) for all pairwise cohort comparisons. These |SMD| values, along with the corresponding P-values, are reported in Supplementary Table 1, whose caption now explains the interpretation of |SMD| and the tests used (Supplementary Table 1). In the 'Participant characteristics of each dataset' subsection of the Results section, we now interpret these statistics, highlighting that External Test 1 differs notably in sex distribution and HDL cholesterol, while External Test 2 is older and shows modest differences in the blood pressure and lipid profiles. We explicitly state that these findings indicate non-trivial clinical heterogeneity between the internal-validation and external cohorts (page 15, lines 326–340).

Reviewer 2

Comments	Responses
[#1. "Consider adding a definition of retinal age early in the abstract, as readers may not be familiar with this concept."]	We agree and have revised the abstract to define retinal age at the very beginning. It now reads: 'Accurate estimation of the retinal age, defined as the age predicted from fundus photographs by a deep-learning model trained on chronological age, provides a non-invasive biomarker of biological ageing and disease risk'. (Abstract, page 3, lines 32–34).
[#2. "The phrase accelerated biological ageing may be misleading; please revise to biological ageing to ensure accuracy."]	Thank you for this important clarification. We have removed the expression 'accelerated biological ageing' from the abstract and now describe retinal age more neutrally as 'a non-invasive biomarker for biological ageing and disease risk' (Abstract, page 3, lines 32–34). In addition, when interpreting the retinal age gap (RAG) we now refer to "older-appearing retinas" rather than accelerated biological ageing: '...indicating older-appearing retinas and supporting the biological relevance of retinal age' (Abstract, page 3, lines 46–49). We emphasise In the Discussion section that retinal age should be interpreted as one component of a broader risk profile rather than a direct measure of organism-wide ageing (Discussion, page 25, lines 557–559).
[#3. "For the additional 7,193 images, clarify whether this cohort was disease-free."]	We have clarified that the 7,288 additional images used for internal validation were also obtained from disease-free adults. The abstract now states that both the training and internal-validation cohorts comprised disease-free individuals: '...50,595 quality-controlled fundus photographs from 27,214 disease-free adults and validated it on an independent set of 7,288 additional images from disease-free adults'. (Abstract, page 3, lines 37–38). In the Methods, we further explain that identical exclusion criteria (history of cardiovascular/cerebrovascular disease, use of antihypertensive, lipid-lowering or

	hypoglycaemic agents, and abnormal cardiometabolic parameters) were applied to both the training and internal-validation cohorts (Methods – Study population and data collection, page 7, lines 129–135), and that 27,214 individuals and a non-overlapping set of 7,288 individuals were then assigned to the training and internal-validation cohorts, respectively (page 7, lines 136–138).
[#4. "The statement about outperforming a public benchmark is confusing. Please specify the source of the benchmark and consider moving this point to the Discussion."]	We greatly appreciate your valuable comment. To avoid confusion, we have removed the phrase 'outperforming a public benchmark' from the abstract. Instead, we now specify the benchmark explicitly in the Results and Discussion sections as the Japan Ocular Imaging Registry (JOIR) retinal age model. In External Test 1, the Results section now states: 'Across all eyes, the MTL ensemble achieved an MAE of 3.39 ± 2.74 years, outperforming the Japan Ocular Imaging Registry (JOIR) retinal age model (5.37 ± 3.84 years), fine-tuned RETFound-DINOv2 model (4.45 ± 3.15 years), and RETFound-DINOv2 baseline (4.65 ± 3.31 years; all P ≤ 0.003)'. (Results – External validation and clinical associations, page 18-19, lines 413–421). In the Discussion, we further contextualise this by explaining that JOIR is a national real-world database and explicitly referring to it as the 'JOIR retinal age benchmark' citing reference [32] (Discussion, page 22, lines 474–477).
[#5. "Clarify why the standard deviation across ensemble predictions was used as an internal confidence score. What was this for?"]	We agree that the rationale should be explicit. In the Results section, we now explain that the standard deviation (SD) across the five-fold-specific age predictions quantifies model disagreement and thus serves as an internal proxy for prediction uncertainty: 'Because this SD reflects the degree of disagreement among the five-fold-specific models and thus serves as an internal measure of prediction uncertainty, we divided eyes into Low-SD (higher confidence; SD below the cohort median of 1.71 years) and High-SD (lower confidence;

	SD at or above the median) subgroups'. (Results – Stratification by prediction standard deviation, page 18, lines 401–406). In the Methods we state that, for each eye, 'we also calculated the SD of the five-fold-specific age predictions as an internal proxy for prediction uncertainty, with larger SD indicating greater disagreement among models. This SD was used to stratify eyes into Low-SD and High-SD subgroups at the cohort-specific median in subsequent accuracy and systemic-disease analyses'. (Methods – Machine learning methods, page 11, lines 232–236). We then used these subgroups consistently in the internal- and external-validation analyses (Tables 3–5) and in the systemic disease associations (Table 6).
[#6. "Only age and sex were adjusted for in the systemic disease cohort; potential confounding factors such as comorbidities and medication use should be acknowledged, as they may bias the results."]	We greatly appreciate this important point. In the Methods we now explicitly state that propensity-score matching in the systemic disease assessment set was based only on age and sex (Methods – Statistics and reproducibility, page 15, lines 318–321). In the Discussion we acknowledge the resulting residual confounding more clearly: 'Propensity-score matching was performed only on age and sex; thus, residual confounding by comorbidities, overall medication burden, and lifestyle factors may have biased the observed associations between retinal age gap and systemic conditions'. (Discussion, page 25, lines 554–556).
[#7. "The statement 'Retinal imaging techniques … provide insights into the health of the eye and systemic circulation' requires citations. Suggested references: Oculomics: Current Concepts and Evidence / Oculomics: A Crusade Against the Four Horsemen of Chronic Disease."]	Thank you for your helpful suggestions. We now explicitly cite oculomics frameworks in support of this statement in the Introduction section: 'Retinal imaging techniques such as fundus photography and optical coherence tomography (OCT) provide a non-invasive view of ocular and systemic circulation, as highlighted in oculomics frameworks that link ocular findings to systemic disease [3–5]'. (Introduction, page 5, lines 68–71). References [3–5]

	include Wagner et al. (oculomics overview), Zhu et al. 'Oculomics: current concepts and evidence', and Patterson et al. 'Oculomics: a crusade against the four horsemen of chronic disease'.
[#8. "The section on AI-based disease detection should cite landmark studies in the field, e.g.: Diabetic retinopathy etc."]	We have expanded the Introduction to cite landmark AI studies in ophthalmic disease detection. Immediately following the oculomics sentence, we now state: 'Deep-learning systems are able to detect major eye diseases, including diabetic retinopathy, glaucoma, and age-related macular degeneration, with high accuracy in large multiethnic cohorts, as illustrated by models for diabetic retinopathy screening and glaucomatous optic neuropathy [6–10]'. (Introduction, page 5, lines 71–74). These references now include Gulshan et al. (JAMA 2016) and Ting et al. (JAMA 2017) for diabetic retinopathy, as well as Li et al. (Ophthalmology 2018) for glaucomatous optic neuropathy, in addition to relevant work on AMD and small-data transfer learning.
[#9. "The concept of RAG and its association with cardiovascular disease should be supported by key references… stroke risk / arterial stiffness / incident CVD."]	We have revised the Introduction to more fully reference the cardiovascular literature on the retinal age gap (RAG). After introducing Poplin et al.'s age-prediction model, we now write: 'The difference between retinal-predicted and chronological age—often termed the retinal age gap (RAG)—has been associated with cardiovascular disorders, including stroke and incident cardiovascular disease mediated by increased arterial stiffness [11–16]'. (Introduction, page 5, lines 77–80). References [15–16] specifically cover stroke risk and associations with arterial stiffness and incident cardiovascular disease.

[#10. "Predictive potential of retinal age… COPD risk / morbidity & mortality / kidney failure / Parkinson's… include landmark studies."]	We have expanded the same paragraph to include these outcomes and their key references. The Introduction now continues: 'Large cohort studies have further demonstrated that retinal ageing markers, including RAG and related indices, predict a wide range of outcomes such as chronic obstructive pulmonary disease, all-cause morbidity and mortality, kidney failure and Parkinson's disease, supporting their potential role as general indicators of systemic ageing and vulnerability [13,14,17–19]'. (Introduction, page 5, lines 80–84). References [13,14,17–19] include retinal age associations with morbidity/mortality, COPD, kidney failure, and Parkinson's disease.
[#11. "The scoping review on retinal age estimation… should cite: Ocular Ageing Biomarkers… Cross-Population…"]	We have updated the section on existing retinal age models to incorporate these additional reviews. The Introduction now states: 'A recent scoping review of retinal age estimation studies published between 2022 and 2023 reported mean absolute errors (MAEs) ranging from 3.30 to 3.97 years [20], while complementary work has summarised ocular ageing biomarkers and cross-population retinal ageing models [21,22]'. (Introduction, page 5, lines 85–88). References [21] and [22] correspond to the suggested reviews on ocular ageing biomarkers and cross-population retinal ageing models.
[#12. "Clarify why only one systemic biomarker (HbA1c) was chosen although deployment uses only fundus photographs."]	We greatly appreciate the opportunity to clarify this design choice. In the Introduction we now explain that we intentionally restricted the multitask learning (MTL) architecture to two outputs—age and a single systemic biomarker—to maintain architectural simplicity, avoid negative transfer from heterogeneous tasks, and isolate the effect of adding one auxiliary task: 'We restricted the MTL architecture to two regression outputs, age and exactly one systemic biomarker, to maintain architectural simplicity, facilitate interpretation of the effect of adding a single auxiliary

	prediction task, and reduce the risk of negative transfer from multiple heterogeneous tasks'. (Introduction, page 6, lines 102–105). We then state that among candidate auxiliary outputs (HbA1c, systolic blood pressure, LDL cholesterol), we selected HbA1c based on empirical performance and its biological properties: '...we ultimately selected HbA1c because the age–HbA1c configuration yielded the lowest age-prediction error , in comparative experiments and because HbA1c is a clinically established and temporally stable marker of metabolic status'. (Introduction, page 6, lines 102–106; Table 2, page 40). Importantly, we also clarify that systemic measurements are used only during training: 'The deployed model requires only a single fundus photograph at inference, as systemic measurements are used solely as auxiliary training signals, enabling an image-only workflow compatible with routine ocular imaging and scalable screening'. (Introduction, page 6, lines 107–109). We further justify HbA1c as a stable supervisory signal and summarise supporting epidemiological evidence in the Discussion section (Discussion, page 23, lines 502–512).
[#13. "In MTL with two outputs, only one MAE is reported. Clarify whether this MAE refers to age only, or how outputs were integrated."]	We have clarified throughout that the reported performance metrics for the MTL models refer to the age-prediction output only. In the Results describing the model-comparison experiments we now explicitly state: 'Table 2 presents a comparison of age-prediction accuracy across various single-task learning (STL) and MTL models; all metrics are computed for the age output in each configuration'. (Results – Performance of single-task and multitask models, page 16, lines 353–355). The footnote to Table 2 has been revised to read: 'For multitask learning (MTL) models, performance metrics refer to the age-prediction output; the auxiliary systemic variable (for example, HbA1c) was not used as a primary selection target'.

	(Table 2 footnote, page 40, lines 874–885). We do not integrate outputs when computing MAE, MSE, or MAPE; these metrics are always calculated for age alone.
[#14. "Please include the age range of participants in each dataset."]	We have added the age range (minimum–maximum) for each cohort to Table 1. The row 'Age, years' in Table 1 now reports both the mean ± SD and the range, for example: '52.3 ± 14.8 (17–94)' for the training cohort, '50.3 ± 15.2 (16–91)' for the internal-validation cohort, '53.1 ± 9.6 (25–69)' for External Test 1, and '57.0 ± 12.8 (30–95)' for External Test 2 (Table 1, page 39, lines 859–869).
[#15. "Cross-validation may risk overfitting; consider addressing this limitation."]	We agree and now explicitly acknowledge this limitation in the Discussion. We state: 'Model selection and hyperparameter tuning were based on cross-validation within a single institutional cohort; thus, the ensemble may still be partially overfitted to this source population despite early stopping and regularisation. Larger, fully independent and ethnically diverse external cohorts with longitudinal follow-up are warranted to more definitively establish the generalisability and prognostic value'. (Discussion, page 25, lines 564–568).
[#16. "The Grad-CAM results need clearer explanation and interpretation."]	Thank you very much for your valuable comment. We have clarified both the technical description of the Grad-CAM visualisations and their interpretation. First, we now describe more explicitly in the Methods how the Grad-CAM maps were generated and displayed. Specifically, we state that Grad-CAM was computed from the last convolutional block and that the resulting maps were min–max normalised on a per-image basis, mapped to a colour scale in which warmer colours indicate relatively higher contribution, and overlaid semi-transparently on the original fundus photograph; we also emphasise that these maps are used as qualitative saliency visualisations rather than

	quantitative estimates of feature importance (Methods – Machine learning methods, page 12, lines 254–264). Second, we have added a brief interpretation of the Grad-CAM patterns to the Discussion. We now note that Grad-CAM visualisations indicate that both the fold-specific models and the ensemble primarily attend to the optic disc, macula, and major temporal vascular arcades while down-weighting peripheral artefacts such as image borders (Fig. 3), which support the biological plausibility of the learned features in the context of retinal ageing (Discussion, page 21, lines 458–462).
[#17. "Rationale for stratifying by prediction SD should be explicitly stated (same as above)."]	As noted in our response to Comment #5, we now explain explicitly that the SD across fold-specific predictions is used as an internal measure of prediction uncertainty and the basis for stratification. In the Results we state: 'Because this SD reflects the degree of disagreement among the five fold-specific models and thus serves as an internal measure of prediction uncertainty, we divided eyes into Low-SD (higher confidence; SD below the cohort median of 1.71 years) and High-SD (lower confidence; SD at or above the median) subgroups'. (Results, page 18, lines 401–406). In the Methods we further note that this SD 'was used to stratify eyes into Low-SD and High-SD subgroups at the cohort-specific median in subsequent accuracy and systemic-disease analyses'. (Methods – Machine learning methods, page 11, lines 232–234). Finally, in the Discussion we highlight that this stratification appears to identify subgroups in which RAG is more informative: 'Hypertension did not reach significance in the overall cohort; however, it was significant in the subset with low ensemble SD, suggesting that stratifying eyes by ensemble disagreement can serve as a proxy for prediction confidence and may help identify subgroups in which retinal age is most informative'. (Discussion, page 24, lines 535–538).

[#18. “Clarify whether the ‘non-medication’ group includes only individuals without hypertension, or also untreated hypertensives. Same for diabetes and lipid-lowering medications.”]	We agree that this distinction should be explicit. In the Methods we have added a sentence clarifying the composition of the non-medication group: ‘For each drug class (antihypertensive, hypoglycaemic, or lipid-lowering agents), ‘Yes’ indicated current use of the corresponding medication, whereas ‘No’ indicated no such reported use; therefore, the non-medication group may include both disease-free individuals and those with untreated or undiagnosed conditions’. (Methods – Measurement of clinical variables, page 10, lines 205–208). This applies equally to hypertension, diabetes and lipid-lowering medications, and we have interpreted the systemic disease results with this limitation in mind.
[#19. “The discussion section is too long and should be more concise.”]	Thank you for your valuable suggestion. We have reorganised the Discussion to improve clarity and conciseness. The first paragraph now succinctly summarises the main methodological contribution and key findings (Discussion, page 21, lines 449–462). Subsequent paragraphs are grouped into: (1) context within retinal AI and foundation models (page 21–22, lines 463–495), (2) rationale for the MTL design and the choice of HbA1c (page 22–23, lines 496–515), (3) integration with retinal-systemic disease literature and oculomics (page 23–24, lines 516–530), and (4) a consolidated limitations paragraph followed by a short paragraph on future directions and clinical implications (page 24–26, lines 543–583). We believe this structure reduces redundancy and makes the Discussion more focused and easier to read.
[#20. “Absence of prospective validation should be emphasized.”]	We completely agree and have strengthened this point in the limitations paragraph. We now state: ‘Furthermore, although we evaluated the ensemble in two independent external cohorts totalling more than five thousand individuals,

	both cohorts comprised only Asian participants, and we did not evaluate performance in non-Asian or broadly multiethnic biobank populations; in contrast, previous retinal age studies have leveraged biobank-scale datasets to demonstrate prospective associations with incident cardiovascular events and mortality [13,14,17–19]. Model selection and hyperparameter tuning were based on cross-validation within a single institutional cohort; thus, the ensemble may still be partially overfitted to this source population despite early stopping and regularisation. Larger, fully independent and ethnically diverse external cohorts with longitudinal follow-up are warranted to more definitively establish the generalisability and prognostic value'. (Discussion, page 25, lines 559–568). Later in the Discussion we also note that 'Prospective studies are warranted to define clinically meaningful thresholds, determine whether acting on retinal age gap-based alerts improves outcomes...' (Discussion, page 26, lines 576–578).
[#21. "As the authors mentioned a subset with three consecutive years of retinal images… please discuss potential future research of longitudinal validation."]	Thank you for highlighting this important opportunity. In the Methods, we describe the identification of a 3-year longitudinal cohort of 10,356 individuals (35,998 eye records) and note that 'This cohort will be used in subsequent work to examine how the retinal age gap (RAG; predicted minus chronological age) evolves over time and how temporal changes relate to the progression of systemic disease'. (Methods – Study population and data collection, page 7, lines 126–128). In the Discussion, we now explicitly link this cohort to future longitudinal validation: 'In future work, we plan to leverage the 3-year longitudinal cohort described in this study to examine how retinal age gap trajectories relate to incident systemic disease'. (Discussion, page 25, lines 568–570). We hope this clarifies how we intend to address longitudinal and prospective validation in subsequent studies.

<table>
<tr><th colspan="2">Reviewer 3</th></tr>
<tr><th>Comments</th><th>Responses</th></tr>
<tr><td>[#1. The proposed multitask-learning (MTL) framework appears incremental: multitask prediction of age and a systemic biomarker is a relatively straightforward extension of existing models. The improvement in MAE is modest. Please justify the choice of age + HbA1c as the auxiliary task, explain why you did not explore more complex multitask configurations, and clarify what is conceptually new here.]</td><td>We thank the Reviewer for this thoughtful critique. We agree that MTL as a general concept is well established, and our goal was not to propose an entirely new learning paradigm but to systematically evaluate simple, interpretable configurations and demonstrate their impact on retinal age prediction.
In the Introduction, we now explicitly state this design philosophy. Rather than introducing a new construct of retinal age, we focus on ‘simple architectural choices: a two-output multitask learning (MTL) framework in which age and a systemic biomarker are predicted jointly, and an ensemble of fold-specific models to stabilise predictions’ (page 6, lines 97–100). We further explain that we restricted the MTL architecture to two regression outputs to maintain architectural simplicity, facilitate interpretation of the effect of adding a single auxiliary prediction task, and reduce the risk of negative transfer from multiple heterogeneous tasks (page 6, lines 102–105).
We now justify the choice of HbA1c as the auxiliary task by reporting that we systematically compared candidate auxiliary outputs (HbA1c, systolic blood pressure, and LDL cholesterol) and found that the fundus-only age + HbA1c configuration achieved the lowest age-prediction error among all candidates, with an ensemble MAE of 4.86 years and a statistically significant improvement over the baseline STL model (ΔMAE −0.18 years; $P < 0.001$) (Results, page 16-17, lines 364–374; Table 2, page 40). We also emphasise that HbA1c is a clinically established and temporally stable marker of metabolic status (Introduction, page 6, lines 102–106; Discussion, page 23, lines 499–508).
Conceptually, we now frame our contribution as providing: (1) a transparent, carefully benchmarked demonstration that a simple, two-output MTL configuration</td></tr>
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	can achieve MAEs below 3 years in internal validation and outperform both a widely used retinal age model and state-of-the-art foundation models in external datasets; and (2) an explicit comparison showing that more complex MTL variants (for example, predicting age with systolic blood pressure or LDL cholesterol) can degrade performance, underscoring the importance of task choice and negative transfer. These points are described in the Results section (page 16-17, lines 353–373, 377–383) and interpreted in the Discussion section (page 21–24, lines 449–530).
[#2. The clinical impact of the observed retinal age gap (RAG) differences is not clearly discussed. The absolute RAG differences appear small; what do these numbers mean in practice, and how might the model be used clinically?]	We greatly appreciate this comment and have added the corresponding text to address the clinical interpretation of RAG magnitude and potential use cases. In the Discussion, we now explicitly interpret the absolute RAG differences observed in our systemic disease cohort, noting that although the differences are measured in years, they are clinically non-trivial when considered at the population level and in relation to prior work (page 24, lines 531–540). We have cited previous studies linking larger retinal age gaps to increased risks of stroke, arterial stiffness, and incident cardiovascular disease, and we emphasise that our findings are directionally consistent with these reports (page 24, lines 538–540). We have also added a paragraph outlining a concrete clinical use case. We propose that, in health-check settings where fundus photography is already performed, individuals with markedly positive RAG values and low ensemble SD could be flagged for more detailed cardiometabolic assessment, whereas near-zero RAG in the context of normal systemic risk factors might provide additional reassurance regarding vascular ageing (Discussion, page 26, lines 573–579). We stress that prospective studies are warranted to define clinically meaningful thresholds and determine whether acting on RAG-based alerts improves outcomes (page 26, lines 576–578).

	Finally, we position retinal age as one component of a broader risk stratification framework that could complement multimodal retinal-phenotype indices such as RetiAGE, rather than as a stand-alone diagnostic (Discussion, page 26, lines 578–579).
[#3. Technical details lack clarity: the QC pipeline is described only briefly; the way non-image inputs are fused with image features is unclear; and the backbone choice (LWBNA_Unet) is not well motivated. Please provide more detail so that the model can be understood and, in principle, reproduced.]	Thank you for your detailed technical feedback. We have enriched the Methods to better describe the QC process, input fusion, and model architecture. Regarding QC, as noted in our response to Reviewer 1, we now describe the expert-led QC procedure in the 'Study population and data collection' subsection, including specific criteria for deeming images poor quality (page 8, lines 158–162). We also explicitly acknowledge residual variation in focus and illumination as a limitation (Discussion, page 24–25, lines 543–552). For input fusion, we now explain in the 'Machine learning methods' subsection that for STL configurations using additional non-image inputs ("Fundus + HbA1c" or "Fundus + Sex"), these covariates were standardised and concatenated with the convolutional feature vector before the fully connected regression head, enabling the network to jointly exploit retinal and systemic information (page 11-12, lines 237–244). Similarly, for MTL configurations, we clarify that the shared convolutional backbone feeds into two parallel fully connected heads corresponding to age and HbA1c outputs (page 12, lines 245–250), and Figure 2 illustrates these architectures. While the full low-level rationale for LWBNA_Unet is detailed in our prior methodological work, in this manuscript, we now summarise that it is a lightweight convolutional backbone with bottleneck and attention modules, chosen to balance accuracy with computational efficiency for large-scale health-check data. We emphasise that our primary focus here is on task configuration (STL vs. MTL, choice of auxiliary target, and ensembling), and we acknowledge in the

	Discussion section that we did not exhaustively compare different backbone architectures or pre-trained models beyond our comparison with RETFound (Discussion, page 25, lines 564–568). This is presented as a limitation and an avenue for future research.
[#4. The manuscript mentions a subset with three consecutive years of retinal imaging, but no longitudinal analysis is presented. Please clarify the role of this cohort and discuss plans for future longitudinal validation.]	We greatly appreciate the opportunity to clarify this point. The 3-year follow-up cohort was constructed as a longitudinal resource for future work rather than being analysed in the present study. In the ‘Study population and data collection’ subsection, we now clearly state that we identified 10,356 individuals with a continuous 3-year follow-up, forming a longitudinal cohort of 35,998 eye records (page 7, lines 125–128). We explicitly note that this cohort ‘will be used in subsequent work to examine how the retinal age gap evolves over time and how temporal changes relate to the progression of systemic disease’ (page 7, lines 126–128). In the Discussion, we have added a statement that larger, ethnically diverse cohorts with longitudinal follow-up are warranted to establish prognostic value and that we plan to leverage the 3-year longitudinal cohort described in this study to examine how RAG trajectories correlate with incident systemic disease (page 25, lines 568–570). We believe this now sets appropriate expectations regarding the current cross-sectional nature of our analyses and our future longitudinal plans.

[#5. The absence of prospective validation and the reliance on self-reported systemic disease status should be more explicitly addressed as limitations. Confounding by comorbidities and lifestyle factors may bias the observed associations.]	We completely agree and have strengthened the limitations section accordingly. In the Discussion, we now clearly state that the systemic disease analyses were based on self-reported diagnoses and medication use, which may contain inaccuracies and do not distinguish, for example, benign arrhythmias from prior myocardial infarction (page 25, lines 552–554). We further acknowledge that propensity-score matching was performed only on age and sex; thus, residual confounding by comorbidities, overall medication burden, and lifestyle factors may have biased the observed associations between RAG and systemic conditions (page 25, lines 554–556). We also emphasise that biological ageing is influenced by genetics, lifestyle, and chronic disease burden, which are not fully captured by retinal imaging alone, and that retinal age should therefore be interpreted as a component of a broader risk profile (page 25, lines 558–560). Finally, we highlight that although we performed external validation in two independent cohorts totalling more than 5,000 individuals, all cohorts were Asian and we did not evaluate the performance in non-Asian or broadly multiethnic biobank populations; in contrast, previous retinal age studies have leveraged biobank-scale datasets with prospective follow-up (page 25, lines 560–564). We state that larger, fully independent and ethnically diverse external cohorts with longitudinal follow-up are warranted to definitively establish generalisability and prognostic value (page 25, lines 566–570).

We hope that these extensive revisions and clarifications fully address the Reviewers' concerns. The manuscript has vastly benefited from your valuable and insightful comments and suggestions. We look forward to hearing from you and would be happy to address any further concerns, if required. We hope this further pushes the manuscript closer to publication in your esteemed journal.